

Starting from McFadden's general GEV theorem, prove that multinomial logit emerges when

$$G(y_1, \dots, y_J) = \sum_{j=1}^J y_j.$$

Then derive the nested logit case for

$$G = \left(\sum_{j \in F} y_j^{\frac{1}{\lambda}} \right)^{\lambda} + \left(\sum_{j \in NF} y_j^{\frac{1}{\lambda}} \right)^{\lambda} + y_0.$$

Verify all four McFadden conditions:

- Non-negativity: $G \geq 0$
- Homogeneity: $G(ty) = t^k G(y)$
- Limiting behavior: $\lim_{y_j \rightarrow \infty} G = \infty$
- Alternating cross-partials: $(-1)^{s+1} \frac{\partial^s G}{\partial y_{i_1} \dots \partial y_{i_s}} \geq 0$

Show the complete derivation of choice probabilities $p_i = \frac{\partial \log G}{\partial u_i}$ for both cases, including all intermediate steps using the chain rule.

Turn in either a snapshot of a handwritten piece of paper, a handwritten PDF from a tablet device, or a compiled LaTeX PDF.

Again, please do not use AI to complete this on your behalf: it robs you of the critical thinking skills you'll develop by doing it "the hard way."